

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI WORK INTEGRATED LEARNING PROGRAMMES

M .Tech (DSE) & M Tech(AIML)

Course Handout(w.e.f. October 2025)

Part A: Content Design

Course Title	Introduction to Statistical Methods
Course No(s)	AIML ZC418 / DSECT ZC418
Credit Units	4
Credit Model	2 - 0.5 - 1.5. 2units for class room hours, 0.5 unit for Tutorial, 1.5 units for Student preparation. 1 unit = 32 hours
Content Authors	Dr YVK Ravi Kumar
Date	October ,2025

Course Description

Basic probability concepts, Conditional probability, Bayes Theorem, Probability distributions, Continuous and discrete distributions, Transformation of random variables, estimating mean, variance, covariance, Hypothesis Testing, Maximum likelihood, ANOVA – single factor, dual factor, time series analysis: AR, MA, ARIMA, SARIMA, sampling based on distribution, statistical significance, Gaussian Mixture Model, Expectation Maximization.

Course Objectives

CO1	Introducing basic concepts of probability and statistics to students
CO2	Students will be able to apply statistical techniques to understand the data
CO3	Students will be able to do statistical analysis of the model / algorithm

Text Books

No	Author(s), Title, Edition, Publishing House
T1	Statistics for Data Scientists, An introduction to probability ,statistics and Data Analysis,Maurits Kaptein et al,Springer 2022
T2	Probability and Statistics for Engineering and Sciences,8 th Edition, Jay L Devore, Cengage Learning
T3	Introduction to Time Series and Forecasting, Second Edition, Peter J Brockwell, Richard A Davis, Springer.

Reference Books

No	Author(s), Title, Edition, Publishing House
R1	Miller and Freund's Probability and statistics for Engineers, 8 th Edition, PHI
R2	Statistics for Business and Economics by Anderson, Sweeney and Wiliams, CENAGE learning

Modular Content Structure

Module 1: Basic Probability & Statistics

- 1.1. Measures of Central Tendency
- 1.2. Measures of Variability
- 1.3. Basic Probability concepts
 - 1.3.1. Axioms of Probability
 - 1.3.2. Definition of Probability
 - 1.3.3. Mutually exclusive and independent events

Module 2: Conditional Probability & Bayes theorem

- 2.1. Conditional Probability
- 2.2. Conditional Probability of Independent events
- 2.3. Bayes Theorem
- 2.4. Introduction to Naïve Bayes concept

Module 3: Probability Distributions

- 3.1. Random Variables
 - 3.1.1. Discrete random variable – Single and two variables
 - 3.1.2. Discrete random variable – Single and two variables
 - 3.1.3. Mean, Variance, Co – Variance of Random variables
 - 3.1.3. Transformation of random variables
- 3.2. Probability Distributions
 - 3.2.1. Bernoulli Distribution
 - 3.2.2. Binomial Distribution
 - 3.2.3. Poisson Distribution
 - 3.2.4. Normal (Gaussian) distribution
 - 3.2.5. Introduction of t – distribution, F – distribution, Chi Square distribution.

Module 4: Hypothesis Testing

- 4.1. Sampling – random sampling and Stratified sampling
- 4.2. Sampling distribution – Central Limit theorem
- 4.3. Estimation – Interval Estimation, Confidence level
- 4.4. Testing of Hypothesis
 - 4.4.1. Mean based
 - 4.4.2. Proportions related
 - 4.4.3. ANOVA – Single and dual factor
- 4.5. Maximum likelihood

Module 5: Prediction & Forecasting

- 5.1. Correlation
- 5.2. Regression
- 5.3. Time Series Analysis
 - 5.3.1. Introduction, Components of time series data
 - 5.3.2. MA model – basic and weighted MA model
 - 5.3.3. Time series models
 - 5.3.3.1. AR Model
 - 5.3.3.2. ARIMA Model
 - 5.3.3.3. SARIMA, SARIMAX, VAR, VARMAX
 - 5.3.3.4. Simple exponential smoothing model

Module 6: Gaussian Mixture model & Expectation Maximization

Learning Outcomes:

No	Learning Outcomes
LO1	Clear understanding of the various statistical models to model the data
LO2	Drawing conclusions from the models selected to understand the data

Part B: Course Handout

Academic Term	I semester ,2025 – 26
Course Title	Introduction to Statistical Methods
Course No	AIML ZC418 / DSECT ZC418

Course Contents

Contact Session 1: Module 1(Module 1:Basic Probability & Statisitcs)

Contact Session	List of Topic Title	Reference
CS - 1	Measures of Central Tendency & Measures of Variability,Data – Symmetric & Asymmetric outlier detection, 5 point summary, Introduction to probability	T1 & T2
HW	Problems on data symmetry & outlier identification	T1 & T2
Lab	Display of statistical Data & Understanding the statistical summary	Lab 1

Contact Session 2: Module 1(Module 1:Basic Probability & Statisitcs)

Contact Session	List of Topic Title	Reference
CS - 2	Axioms of Probability,Mutually exclusive and independent events,Problem solving to understand basic probability concepts	T1 & T2
HW	Problems on probability	T1 & T2
Lab		

Contact Session 3: Module 2(Conditional Probability & Bayes theorem)

Contact Session	List of Topic Title	Reference
CS - 3	Introduction to conditional probability,indepents events, Total probability	T1 & T2
HW	Problems on conditional probability	T1 & T2
Lab		

Contact Session 4: Module 2(Conditional Probability & Bayes theorem)

Contact Session	List of Topic Title	Reference
CS - 4	Bayes theorem(with proof),Introduction to Naïve Bayes concept.	T1 & T2
HW	Problems on Bayes theorem	T1 & T2
Lab	Bayes theorem & Naïve Bayes Concept	Lab 2

Contact Session 5: Module 3: Probability Distributions

Contact Session	List of Topic Title	Reference
CS - 5	Random variables - Discrete & continuous Expectaion of a random variable,mean and variance of a random variable – Sinlge random random variable & Joint distributions	T1 & T2
HW	Problems on random variables	T1 & T2
Lab	Probability Distributions & Sampling	Lab 3

Contact Session 6: Module 3: Probability Distributions

Contact Session	List of Topic Title	Reference
CS - 6	Bernoulli,Binomial, Poisson and Norma distributions. Inroduction to t – distribution , F – Distribution and Chi Sqaure distributions	T1 & T2
HW	Problems on Probability distributions	T1 & T2
Lab	Probability Distributions & Sampling	Lab 3

Contact Session 7: Module 4: Hypothesis Testing

Contact Session	List of Topic Title	Reference
CS - 7	Sampling – random sampling and Stratified sampling,Sampling distribution – Cental Limit theorem,Estimation– Interval Estimation,Confidence level	T1 & T2
HW	Problems on Interval estimation	T1 & T2

Lab		
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Contact Session 8:

Contact Session	List of Topic Title	Reference
CS - 8	REVISION OF THE TOPICS COVERED	
HW		
Lab		

MID SEMESTER EXAMINATION

Contact Session 9 : Module 4: Hypothesis Testing

Contact Session	List of Topic Title	Reference
CS – 9	Testing of Hypothesis - mean and proportions related models (one mean, two mean, one proportion and Several proportions with small and big samples wherever applicable)	T1 & T2
HW	Problems on Testing of Hypothesis	T1 & T2
Lab		

Contact Session 10: Module 4: Hypothesis Testing

Contact Session	List of Topic Title	Reference
CS – 10	Maximum likelihood ,ANOVA – Single and dual factor	T1 & T2
HW	Problems on ANOVA	T1 & T2
Lab	ANOVA	Lab 4

Contact Session 11: Module 5: Prediction & Forecasting

Contact Session	List of Topic Title	Reference
CS - 11	Correlation & regression	T1 & T2
HW	Problems on correlation and regression	T1 & T2
Lab	Regression – Analysis of model summary	Lab 5

Contact Session 12: : Module 5: Prediction & Forecasting

Contact Session	List of Topic Title	Reference
CS - 12	Time Series Analysis - Introduction, Components of time series data, Moving Averages and weighted moving averages model	T3
HW	Problems on Time series	T3
Lab	Time series	Lab 6

Contact Session 13: Module 5: Prediction & Forecasting

Contact Session	List of Topic Title	Reference
CS - 13	Time series models - AR Model, ARMA Model, ARIMA	T3
HW	Problems on Time Series	T3
Lab	Time series	Lab 6

Contact Session 14: Module 5: Prediction & Forecasting

Contact Session	List of Topic Title	Reference
CS - 14	Time Series Models - SARIMA, SARIMAX, VAR, VARMAX and Simple exponential smoothing model	T3

HW	Problems on Time Series	
Lab	Time series	Lab 6

Contact Session 15: Module 6: Gaussian Mixture model & Expectation Maximization

Contact Session	List of Topic Title	Reference
CS - 15	Gaussian Mixture model & Expectation Maximization	Class Notes
HW	Problems on GMM & EM	
Lab		

Contact Session 16:

Contact Session	List of Topic Title	Reference
CS - 16	REVISION OF THE SYLLABUS	
HW		
Lab		

Detailed Plan for Lab work

Lab No.	Lab Objective	Lab Sheet Access URL	Session Reference
1	Display of statistical Data & Understanding the statistical summary		1
2	Bayes theorem & Naïve Bayes Concept		4
3	Probability Distributions & Sampling		5 & 6
4	ANOVA		10
5	Regression – Analysis of model summary		11
6	Time series		12 - 14

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

No	Name	Type	Duration	Weight	Day, Date, Session, Time
EC-1(a)	Quizzes – 1 & 2	Online		10%	
EC-1(b)	Assignments - 2	Online		20%	
EC-2	Mid-Semester Test	Closed Book		30%	
EC-3	Comprehensive Exam	Open Book		40%	

Note:

Syllabus for Mid-Semester Test (Closed Book): Topics in Session Nos. 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics (Session Nos. 1 to 16)